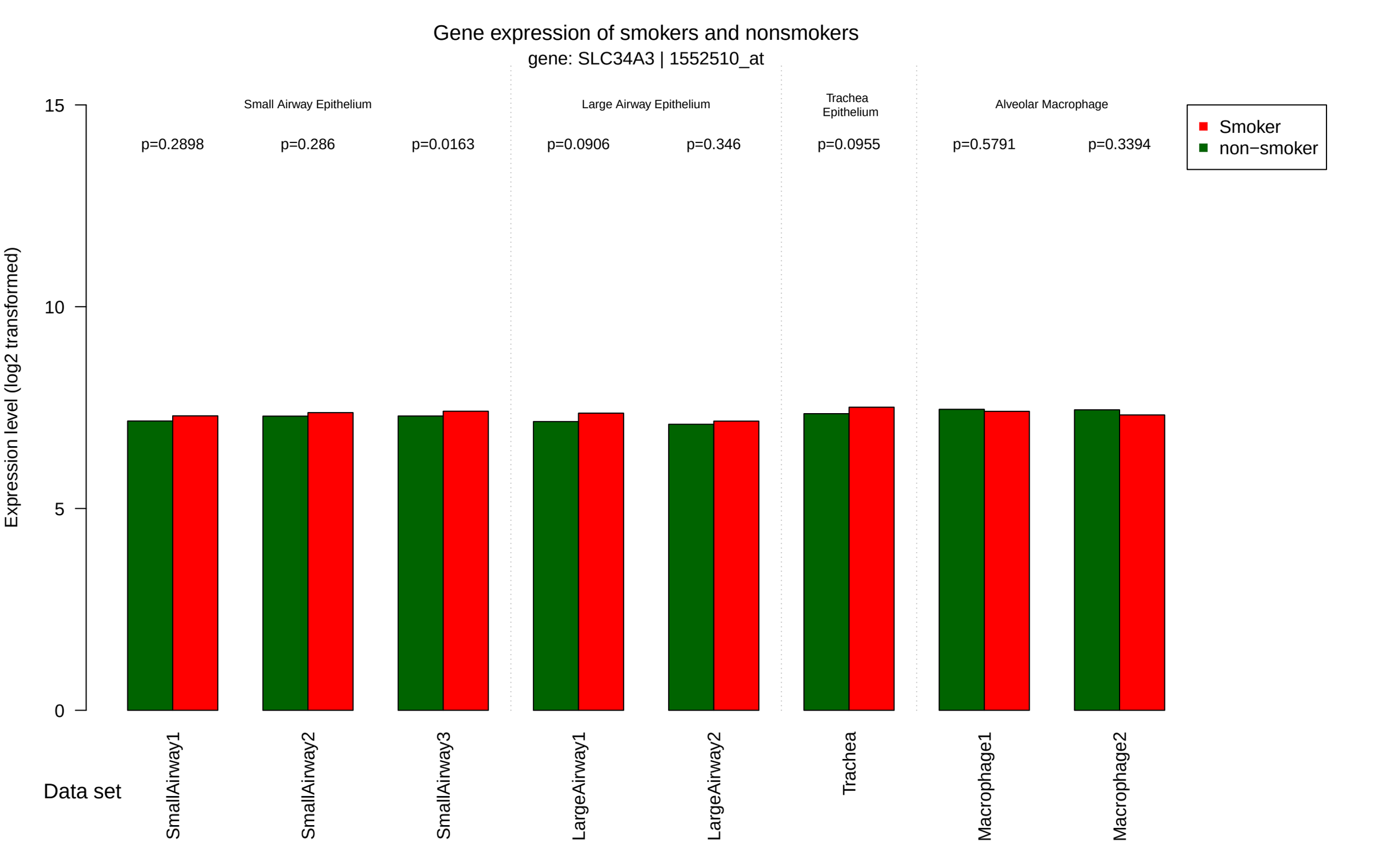
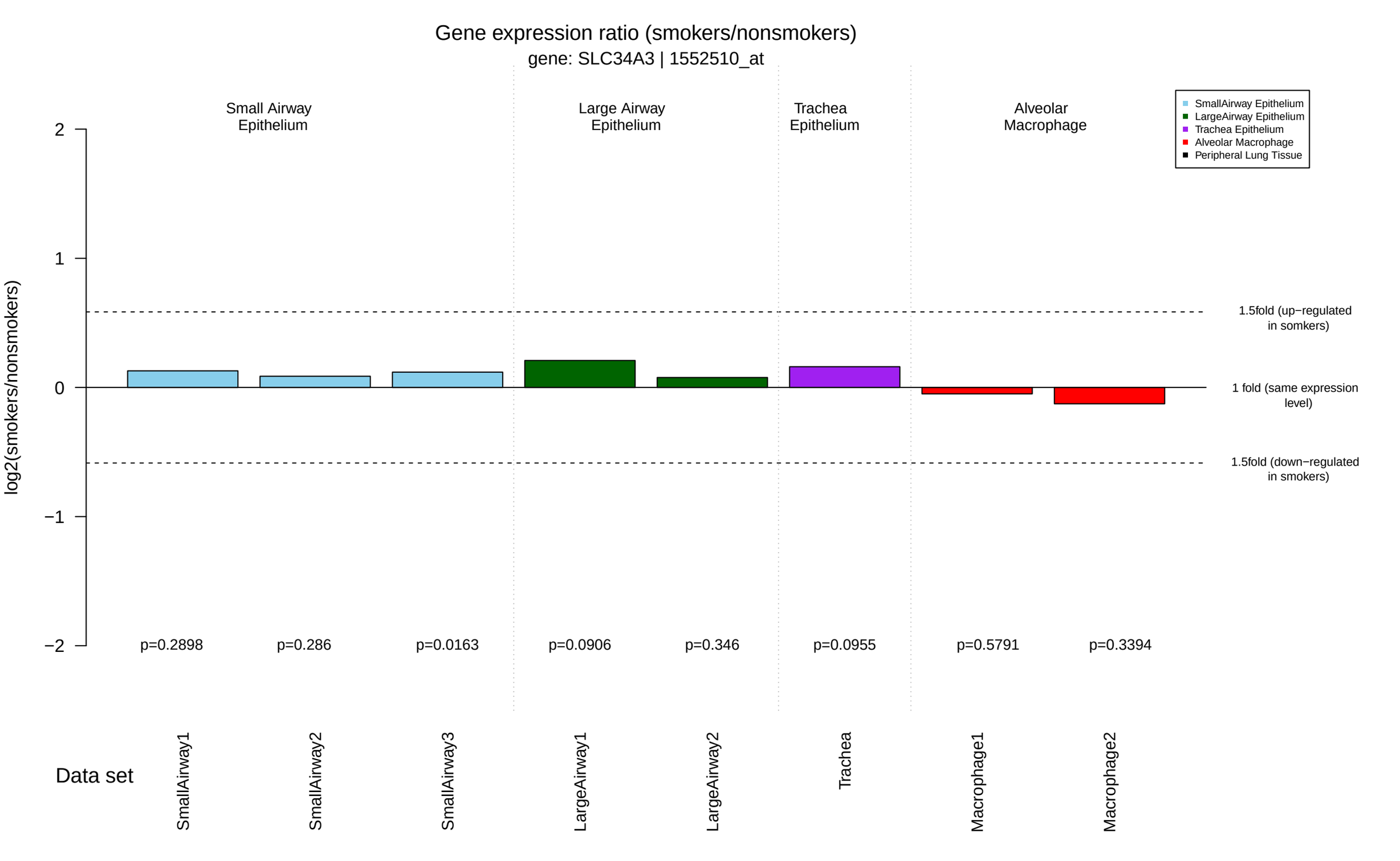
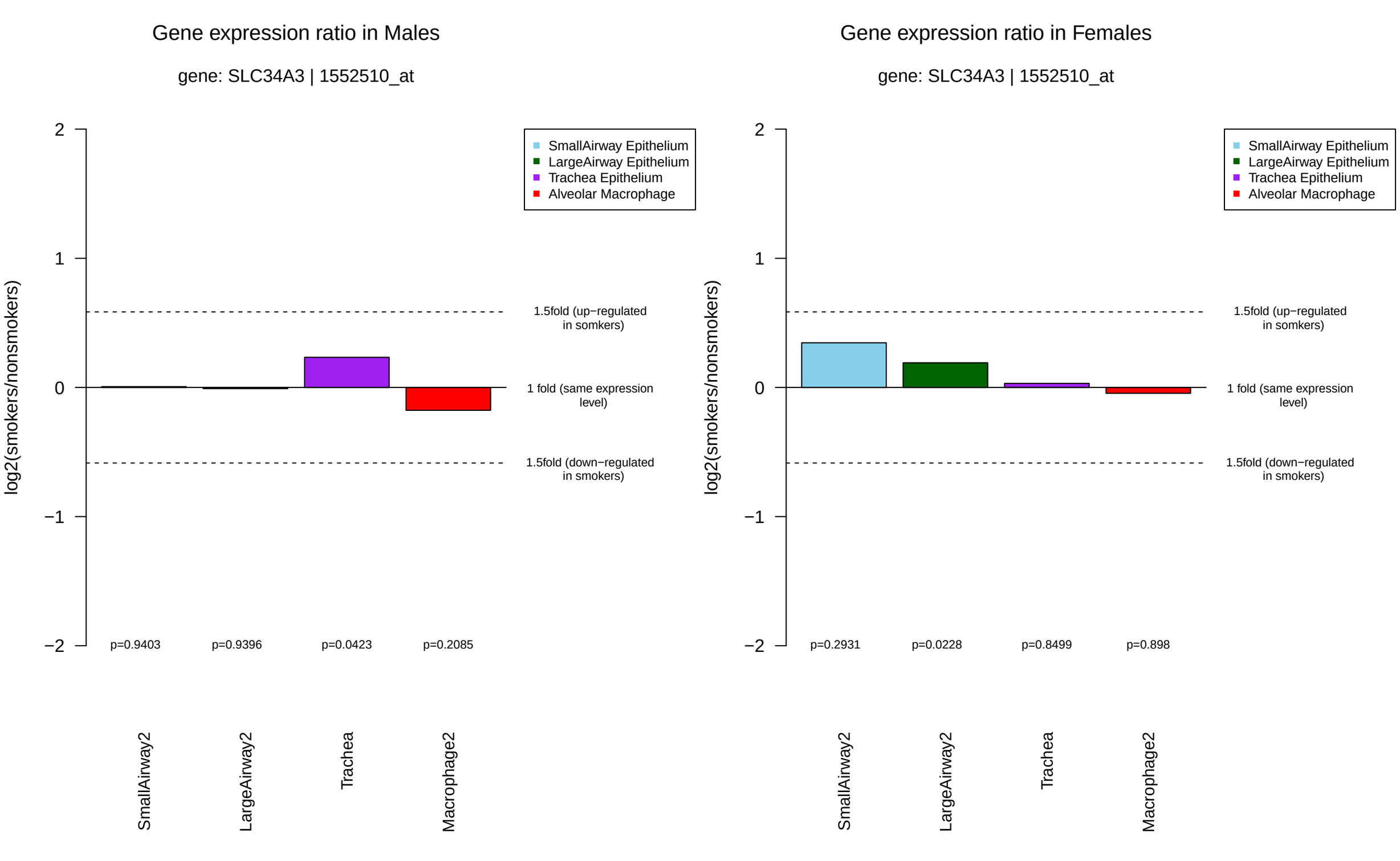
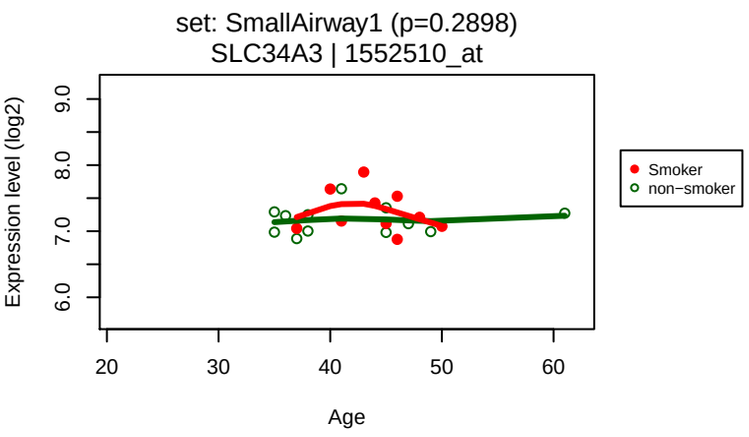
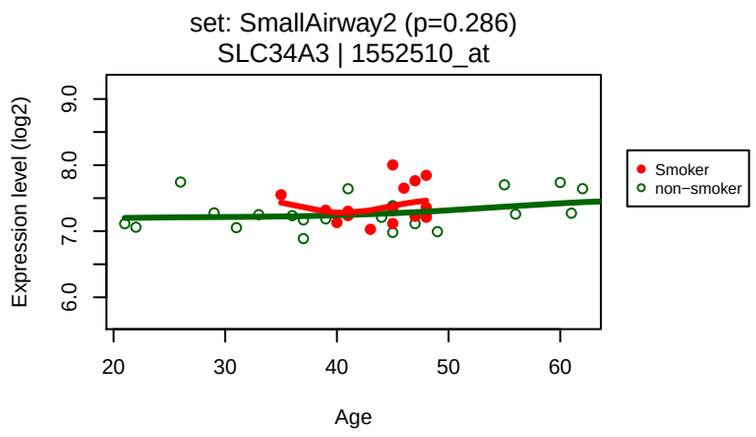
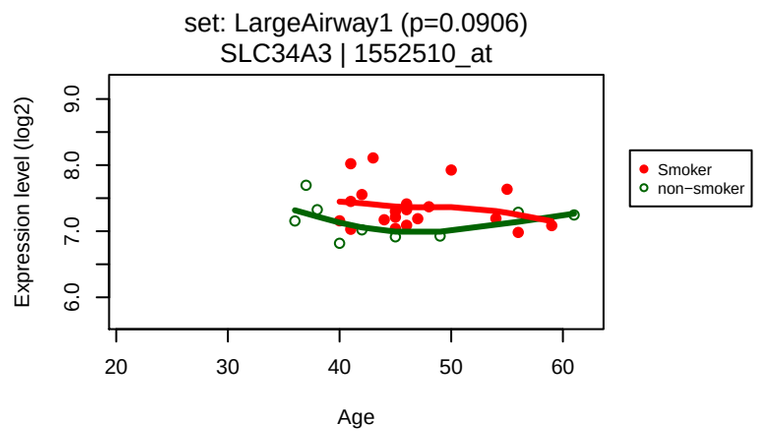

Gene expression of smokers and nonsmokers
gene: SLC34A3 | 1552510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2898
p=0.286
p=0.0163
p=0.0906
p=0.346
p=0.0955
p=0.5791
p=0.3394
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

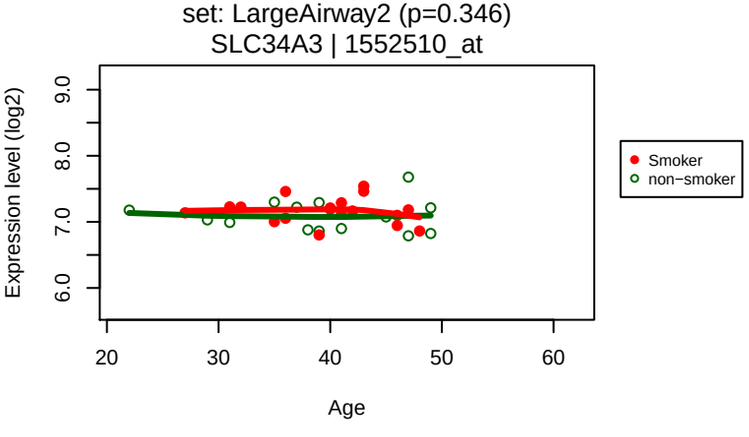

Gene expression ratio (smokers/nonsmokers)
gene: SLC34A3 | 1552510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2898
p=0.286
p=0.0163
p=0.0906
p=0.346
p=0.0955
p=0.5791
p=0.3394
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

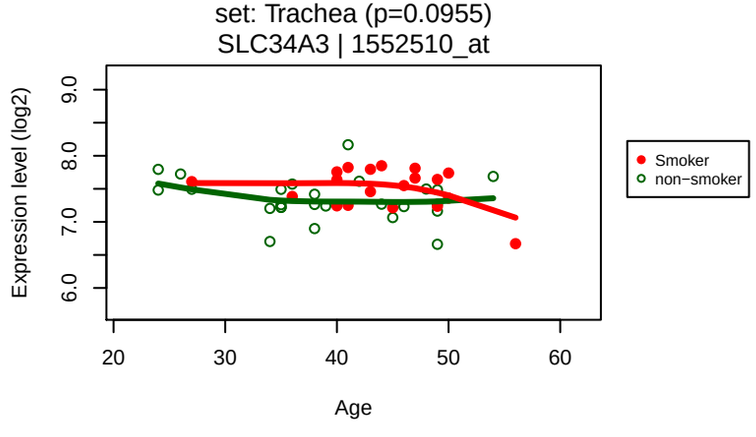

Gene expression ratio in Males
gene: SLC34A3 | 1552510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9403
p=0.9396
p=0.0423
p=0.2085
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC34A3 | 1552510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2931
p=0.0228
p=0.8499
p=0.898
SmallAirway2
LargeAirway2
Trachea
Macrophage2

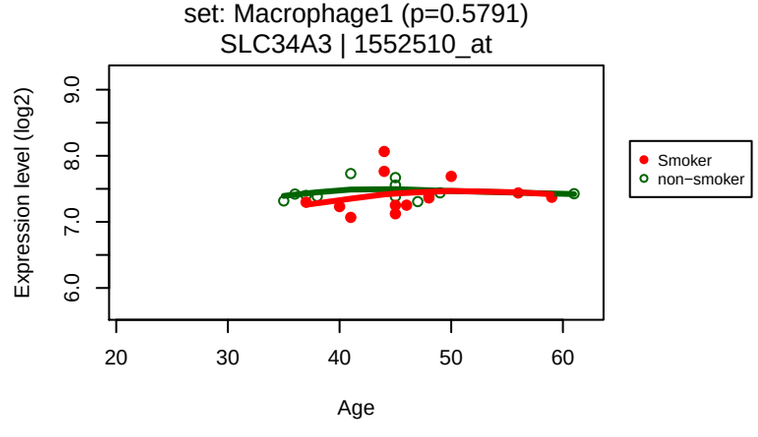

set: SmallAirway1 (p=0.2898)
SLC34A3 | 1552510_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

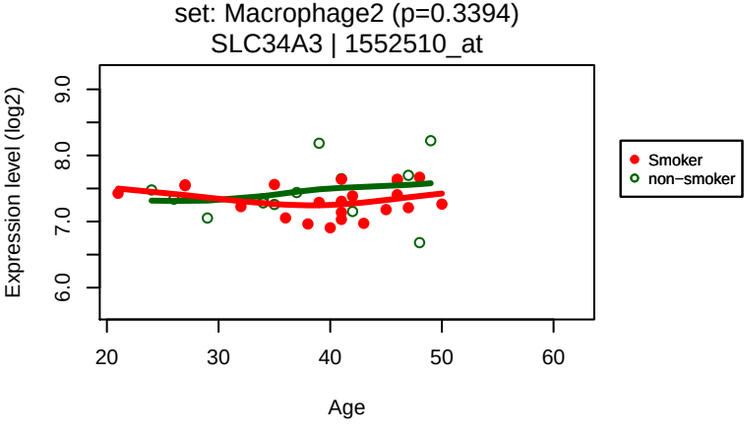

set: SmallAirway2 (p=0.286)
SLC34A3 | 1552510_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

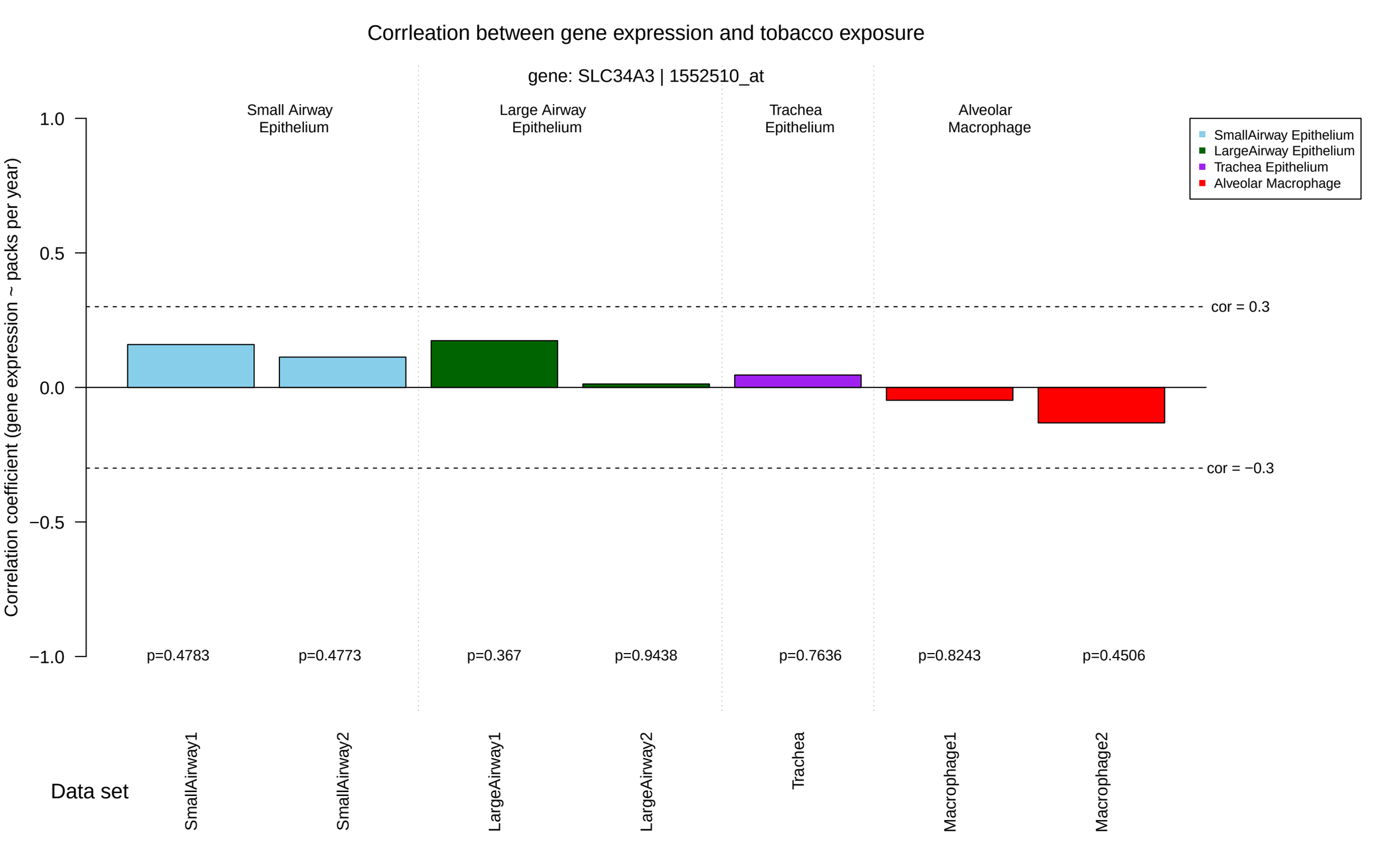

set: LargeAirway1 (p=0.0906)
SLC34A3 | 1552510_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.346)
SLC34A3 | 1552510_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0955)
SLC34A3 | 1552510_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5791)
SLC34A3 | 1552510_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3394)
SLC34A3 | 1552510_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC34A3 | 1552510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4783
p=0.4773
p=0.367
p=0.9438
p=0.7636
p=0.8243
p=0.4506
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2